

LEO ZHOU

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Education

Rice University | B.S. in Computer Science | **GPA: 3.92/4.00** **Expected Graduation May 2028**
• **Relevant Coursework:** Data Structures & Algorithms, Systems Programming, Discrete Math, Linear Algebra

Experience

Software Developer | KEWW **September 2026 – Present**

- Building support, SEO and content agents under human approval; drafts ship unedited 55% and 34 pages went live
- Indexing 210,000 email threads and 15 years of records into Postgres with pgvector, redacting PII with Presidio
- Writing a Kotlin kiosk launcher for 5,000 AOSP smartwatches, relaying tasks over BLE with offline sync
- Fixed a shift double-booking race on a live Laravel platform, then added staging and CI with 40 tests

Undergraduate Research Assistant | Rice University | Prof. Jiarong Xing **August 2026 – Present**

- Ended a forced daily re-login in a macOS usage-tracking app by implementing OAuth token refresh in Swift
- Wrote a Keychain credential inspector after the documented token shape proved incomplete, verified live
- Surfaced usage from Claude Code sessions run over SSH by discovering hosts and mirroring transcripts

Software Engineer Intern | CanSemi **June 2026 – August 2026**

- Parsed 500+ binary test equipment logs in Python into a normalized PostgreSQL schema, 40 lots in under 2 min
- Wrote a FastAPI service with 8 endpoints over that data, cutting report turnaround ~75% for 6 engineers
- Built an interactive React and Canvas wafer map with hover and drill-down, retiring the team's Excel heatmaps

Software Engineer Intern | Inkstone Technologies **April 2026 – June 2026**

- Built a native iOS reading app in Swift and SwiftUI that adapts every screen to the user's known characters
- Wired Apple Vision OCR into a live camera pipeline overlaying pronunciation on unfamiliar characters only
- Parsed and frequency-ranked a 113,000-entry Chinese dictionary in Python into an indexed SQLite database

Embedded Software Engineer | Rice Wind Energy **October 2025 – April 2026**

- Programmed ESP32 firmware in C/C++ reading turbine speed, power output, and blade pitch off live sensors
- Built a six-state safety machine with fault latching and recovery, plus perturb-and-observe power tracking

Projects

BlunderNet | Live Chess Training Platform [🌐 blundernet.com](#) **2026**

Go, AWS, Terraform, Docker, Redis, PostgreSQL, React, PyTorch, C++, Stockfish

- Built and ran a live chess site for ~120 users, filtering 3.25M puzzles by rating, theme, phase and length
- Cut uniform random puzzle selection from 1.4 s to 0.9 ms with a precomputed grid and a stored shuffle key
- Kept the Go API stateless with state in Redis and engine work on SQS workers, all defined in Terraform
- Trained the engine in PyTorch from scratch, rewriting its Monte Carlo tree search in C++ for a 1.8× speedup

Vestigo | Evidence-Grounded Geolocation Agent [🌐 vestigo.earth](#) **Ongoing**

Python, PyTorch, CLIP/SigLIP, LLM agent design, tool calling, evaluation harness

- Built an agent answering at the finest level its evidence supports: observe, guess, use tools, claim, resolve
- Scored claims by granularity instead of distance: 88-90% correct at the level claimed across three photo sets
- Added a solar-position filter cutting worst-case rerun disagreement 14,964 km to 537 km, median unchanged

Technical Skills

Languages: Python, Go, C/C++, Swift, Kotlin, TypeScript/JavaScript, SQL

Backend & Web: FastAPI, Laravel, React, Flutter, SwiftUI, REST APIs, WebSockets, PostgreSQL, Redis, SQLite

Cloud & Infrastructure: AWS (ECS, SQS, RDS), Terraform, Docker, Linux, GitHub Actions, Git

ML & Embedded: PyTorch, LLM agents, pgvector, embeddings, Apple Vision, ESP32, bare-metal C firmware

Leadership & Athletics

- **Junior National Fencing Team** — Team Captain, Coach, Referee; 30+ titles, Junior Worlds + Commonwealth
- **Rice Fencing Club** — Manager · **Physics Pathfinders** — Charity President; 600+ students, 22 schools